Chapter 12 — Asynchronous JavaScript

JavaScript runs your code on **one thread**, yet a web page can download data, animate, and respond to clicks at the same time, and a Node server (Node.js runs JavaScript outside the browser) can handle thousands of connections. The trick is **asynchrony**: slow work is handed to the host (browser or Node), and your code is called back later, when the result is ready. The piece that decides *when* each callback (a function you hand over to be run later) runs is the **event loop**. This chapter explains the event loop precisely — precisely enough to predict the output of tricky ordering puzzles — and then the three generations of async code built on top of it: callbacks, promises, and `async` / `await`.

What you'll learn

1. Why JavaScript is single-threaded and what "blocking" means.
2. The event loop: call stack, task queue, microtask queue, and rendering — with diagrams.
3. Seven ordering puzzles, each run for real in Node **and** Chrome.
4. Node's own additions: `process.nextTick`, `setImmediate`, and the loop phases.
5. Timers: `setTimeout`, `setInterval`, and their guarantees (and non-guarantees).
6. Callbacks, error-first callbacks, and why "callback hell" led to promises.
7. Promises: states, chaining, error flow, `withResolvers`, `Promise.try`.
8. The combinators `all`, `allSettled`, `race`, `any`.
9. `async` / `await`: sequential vs parallel, loops, error handling.
10. Cancellation with `AbortController` and `AbortSignal`.
11. Async iteration: `for await…of`, async generators, `Array.fromAsync`.
12. Production patterns: retry with backoff, concurrency limits, stale responses.

Before you start. You need functions and closures (Chapter 4), and generators/iterators (Chapter 8) for §12.12. Errors and `try` / `catch` are covered in Chapter 11 — §11.5 previewed async errors; this chapter completes the story. Top-level `await` in modules is explained in Chapter 10.

All runnable code is in `code/`. Outputs are real (Node 24). Where timing matters, programs print rounded or yes/no values so the output is stable between runs.

12.1 One thread, and why blocking hurts

What it is

A **thread** is one sequence of instructions being executed. JavaScript gives your code exactly one: at any moment, at most one line of *your* JavaScript is running. **Synchronous** code runs start to finish without pausing. **Asynchronous** code starts an operation now and handles its result later, through a callback.

Analogy: a restaurant with one chef. A good chef doesn't stand watching the oven for 20 minutes — they put the dish in, set a timer (asynchronous), and cook other orders. When the timer rings, they come back to that dish. A bad chef who stares at the oven (blocking) makes every other customer wait.

Terms:

- **Blocking** — code that keeps the thread busy, so nothing else (other callbacks, clicks, rendering — drawing the page on screen) can run.
- **Non-blocking I/O** — input/output (network, disk) performed by the host in the background; JavaScript only runs when the result arrives.

- **Host / runtime** — the environment around the engine: the browser (Web APIs — the extra features a browser provides, such as timers, `fetch` and the page itself) or Node (libuv, its C library for async I/O). The engine (V8, Google's JavaScript engine inside Chrome and Node) runs JavaScript; the host does the waiting.

Why it matters

In a browser, the same thread that runs your JavaScript also handles user input and (usually) page rendering. A 200 ms synchronous loop freezes scrolling, typing and animation for 200 ms. In Node, a blocking handler stalls *every* request the server is handling.

`code/01_single_thread.mjs`:

```
const start = Date.now();
setTimeout(() => {
  const waited = Date.now() - start;
  console.log(`timer asked for 0 ms, ran after >= 200 ms: ${waited >= 200}`);
}, 0);

console.log('busy loop starts');
while (Date.now() - start < 200) { /* burn CPU: nothing else can run */ }
console.log('busy loop ends (the timer is still waiting)');

// Asynchronous work does NOT block: the callback is queued and we carry on.
import { readFile } from 'node:fs/promises';
readFile(new URL(import.meta.url)).then((buf) => console.log('file read finished,', buf.length > 0 ?
'got bytes' : 'empty'));
console.log('readFile started - main script keeps going');
```

Output:

```
busy loop starts
busy loop ends (the timer is still waiting)
readFile started - main script keeps going
timer asked for 0 ms, ran after >= 200 ms: true
file read finished, got bytes
```

The 0 ms timer could not interrupt the loop. JavaScript callbacks are never pre-empted (interrupted part-way through to let something else run): a callback runs to completion before anything else starts (**run-to-completion**). That rule is a gift — you never need locks (guards that stop two threads from changing the same data at the same moment) around your variables — and a responsibility: keep each piece of work short.

When you really need parallel CPU work

Heavy computation (image processing, big parsing) belongs off the main thread (the single thread that runs your page's JavaScript and its UI): **Web Workers** in the browser (Chapter 14) or `worker_threads` in Node (Chapter 15). Workers are separate threads with separate memory that talk by messages.

> **TL;DR**
>
> One thread runs your JavaScript; the host waits for I/O and timers in the background. Long synchronous work blocks everything, including 0 ms timers and rendering.

12.2 The event loop

What it is

The **event loop** is the host's scheduler. It repeatedly picks a piece of waiting work and runs it on the single JavaScript thread. The parts:

Part	What it holds	Examples
Call stack	Functions currently executing, newest on top	Your main script, a callback in progress
Host APIs (Web APIs / libuv)	Operations in progress *outside* JavaScript	A timer counting down, an HTTP request (a message asking a web server for data), a file read
Task queue (a.k.a. macrotask / callback queue)	Callbacks ready to run, one per loop turn	Timer callbacks, I/O callbacks, `message` events, user events like `click`
Microtask queue	Short follow-up jobs that must run "as soon as the current code finishes"	Promise `then` / `catch` / `finally` callbacks, code after `await`, `queueMicrotask`, `MutationObserver` (watches the page for changes) callbacks
Rendering (browsers)	Style (which CSS applies), layout (sizes and positions), paint (drawing pixels) — typically up to ~60 times a second	`requestAnimationFrame` callbacks run just before paint

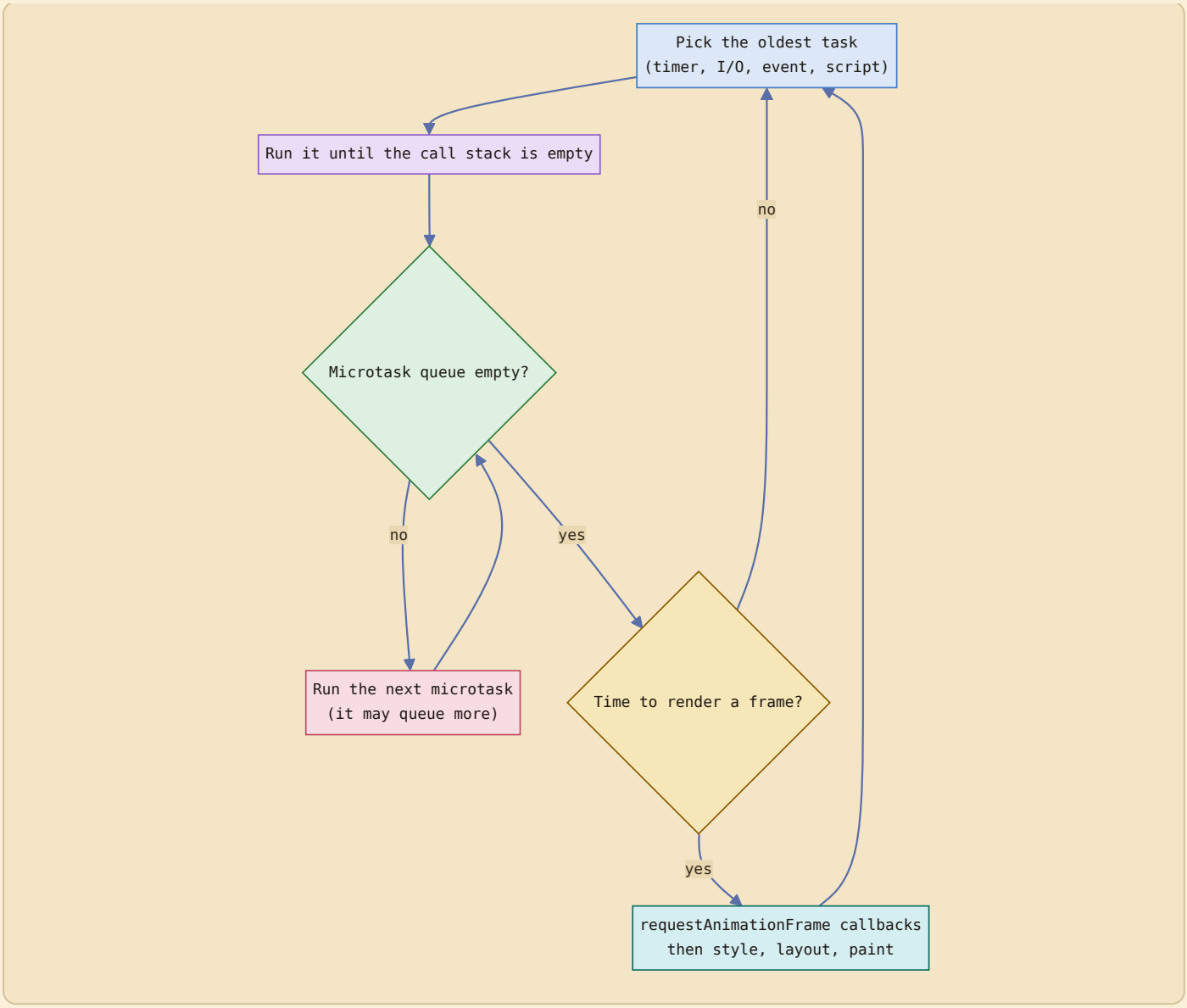
The HTML specification (section "Event loops") defines the browser algorithm; ECMAScript (the official name of the JavaScript language standard) itself only defines **jobs** (promise reactions) and leaves scheduling to the host. The spec actually allows several task queues (for example, user-input tasks may be prioritised over timers), but for predicting order you can treat tasks as one FIFO (first in, first out — like a queue at a shop) queue per source.

How it works

One turn of the browser event loop, simplified from the HTML spec:

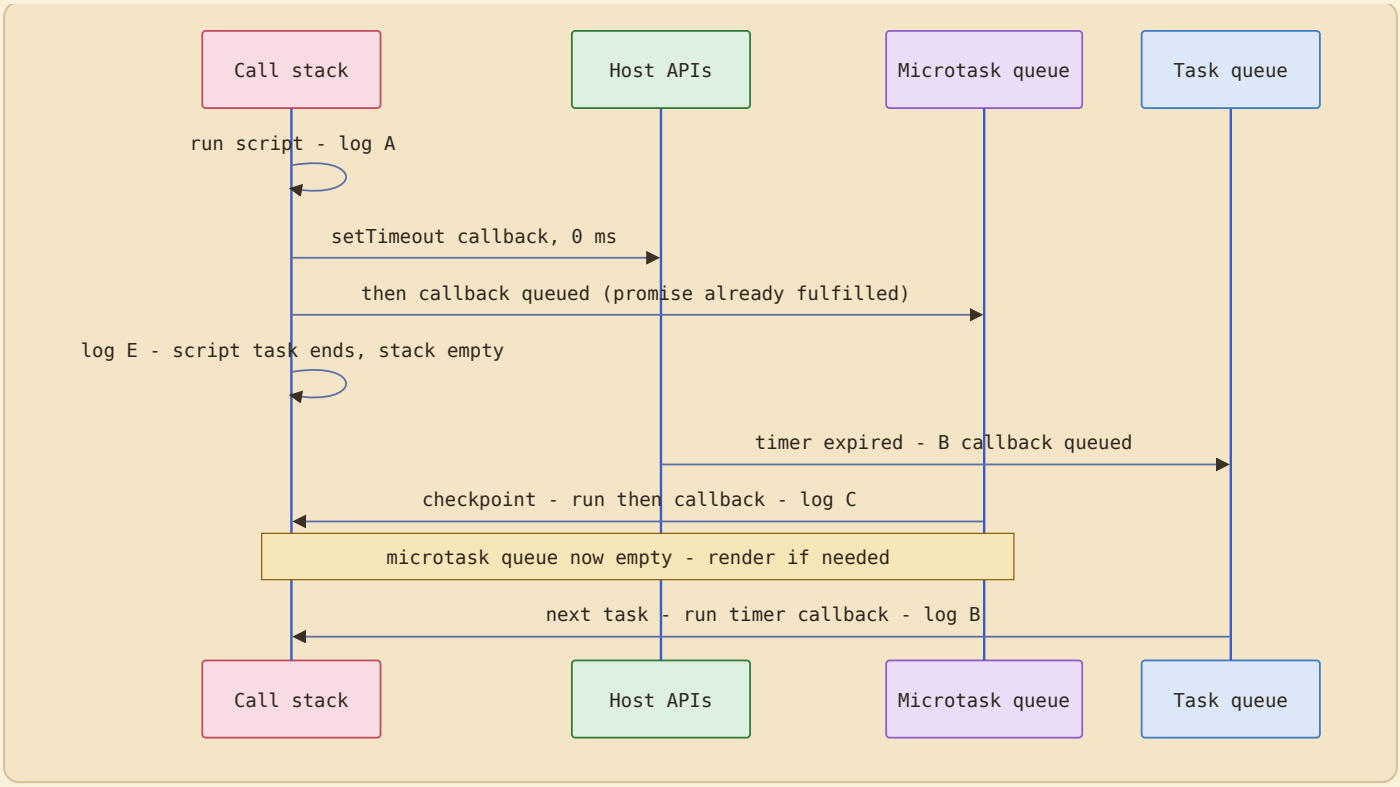
1. **Run one task** — take the oldest runnable task and run its callback until the call stack is empty.
2. **Perform a microtask checkpoint** — run microtasks until the microtask queue is **empty**, including any microtasks queued *by* microtasks.
3. **Maybe render** — if it's time for a new frame (one fresh picture of the page on screen) (and something changed): run `requestAnimationFrame` callbacks, recalculate style and layout, paint.
4. Go to step 1 (or sleep until a task arrives).

The initial script is itself a task, so step 2 happens right after your top-level code finishes. Microtasks also run whenever the call stack becomes empty during a task — for example between two event listeners when the user physically clicks (but not when you call `el.click()` from code, because your code is still on the stack).



The same flow as a sequence over time, for this code:

```
console.log('A');
setTimeout(() => console.log('B timeout'), 0);
Promise.resolve().then(() => console.log('C then'));
console.log('E');
```



Output: `A`, `E`, `C then`, `B timeout`.

Why microtasks exist

A promise callback should run **after** the current synchronous code (so behaviour is consistent — callbacks are always asynchronous) but **before** anything else gets a chance to observe inconsistent state (no other task, no render in between). Microtasks give exactly that "at the end of the current job" timing.

Common mistakes

- **Thinking `setTimeout(fn, 0)` runs "immediately".** It runs in a *later task*, after every pending microtask and possibly after a render.
- **Microtask starvation.** A microtask that always queues another microtask (e.g. a recursive `.then` loop) never lets the loop reach step 3 or step 1 — the page freezes exactly like a `while(true)`. Use `setTimeout` (a task) to yield (hand the thread back so other work can run) when chunking long work (splitting it into small pieces).
- **Expecting `await` to yield to rendering.** `await` only schedules a microtask; the browser can't paint in between. To let the page update during a long job, split it across tasks (`setTimeout`, `scheduler.yield()` where supported, or `requestAnimationFrame`).

TL;DR

Loop: one task → **all** microtasks → maybe render → repeat. Promises and `await` continuations are microtasks; timers, I/O and events are tasks.

12.3 Ordering puzzles

Interviews love "what does this print?" questions. With the rules from §12.2 you can answer any of them:

1. Run all synchronous code first (including promise **executors** — the function you pass to `new Promise` — and code before the first `await`).

2. Then drain the microtask queue in FIFO order, adding new microtasks to the end as they're created.
3. Then take the next task (timers in order of due time; equal times in creation order), run it, drain microtasks again, and so on.

`code/02_ordering_puzzles.mjs` runs seven puzzles. Each one logs into an array, the program waits for all queues to drain, then prints the order. (Try predicting each before reading the answer.)

```
// P1 - the classic: sync code, then ALL microtasks, then the next task
puzzle('P1 classic', (log) => {
  log('A');
  setTimeout(() => log('B timeout'), 0);
  Promise.resolve().then(() => log('C then'));
  queueMicrotask(() => log('D microtask'));
  log('E');
});

// P2 - async/await: code before the first await runs synchronously
puzzle('P2 async/await', (log) => {
  async function f() {
    log('f start');
    await null;             // everything after this line is a microtask
    log('f after await');
  }
  log('1');
  f();
  log('2');
});

// P3 - microtasks queued BY microtasks still run before the next task
puzzle('P3 nested', (log) => {
  setTimeout(() => {
    log('T1');
    Promise.resolve().then(() => log('T1 -> micro'));
  }, 0);
  setTimeout(() => log('T2'), 0);
  Promise.resolve().then(() => {
    log('M1');
    setTimeout(() => log('M1 -> timeout'), 0);
    Promise.resolve().then(() => log('M1 -> micro'));
  });
});

// P4 - two promise chains interleave one step at a time
puzzle('P4 interleave', (log) => {
  Promise.resolve().then(() => log('a1')).then(() => log('a2')).then(() => log('a3'));
  Promise.resolve().then(() => log('b1')).then(() => log('b2')).then(() => log('b3'));
});

// P5 - the executor of new Promise runs synchronously; then() callbacks do not
puzzle('P5 executor', (log) => {
  log('start');
  const p = new Promise((resolve) => {
    log('executor');
    resolve('value');
    log('after resolve');   // resolve() does not stop the function
  });
  p.then((v) => log(`then: ${v}`));
  log('end');
});

// P6 - returning a promise from then() costs extra microtask ticks
puzzle('P6 return promise', (log) => {
  Promise.resolve()
    .then(() => { log('x1'); return Promise.resolve('inner'); }) // adopts: +2 ticks
    .then((v) => log(`x2 (${v})`));
  Promise.resolve()
    .then(() => log('y1'))
    .then(() => log('y2'))
    .then(() => log('y3'))
```

```
    .then(() => log('y4'));
});

// P7 - timers with different delays: order is by due time, then by creation
puzzle('P7 timer delays', (log) => {
  setTimeout(() => log('20ms'), 20);
  setTimeout(() => log('0ms first'), 0);
  setTimeout(() => log('1ms'), 1);
  setTimeout(() => log('0ms second'), 0);
});
// ... the harness runs each puzzle, waits 50 ms, prints the collected order
```

Output (Node 24):

```
P1 classic         -> A | E | C then | D microtask | B timeout
P2 async/await     -> 1 | f start | 2 | f after await
P3 nested          -> M1 | M1 -> micro | T1 | T1 -> micro | T2 | M1 -> timeout
P4 interleave      -> a1 | b1 | a2 | b2 | a3 | b3
P5 executor        -> start | executor | after resolve | end | then: value
P6 return promise  -> x1 | y1 | y2 | y3 | x2 (inner) | y4
P7 timer delays    -> 0ms first | 1ms | 0ms second | 20ms
```

The same puzzles, pasted into a web page (`code/puzzles_browser.html`) and run by headless Chrome (Chrome running without a visible window, driven by a program), give the identical order — promise and timer ordering is standardized, not a Node quirk:

Chapter 12 ordering puzzles — run by headless Chrome

Puzzle	Order the browser produced
P1 classic	A | E | C then | D microtask | B timeout
P2 async/await	1 | f start | 2 | f after await
P3 nested	M1 | M1 -> micro | T1 | T1 -> micro | T2 | M1 -> timeout
P4 interleave	a1 | b1 | a2 | b2 | a3 | b3
P5 executor	start | executor | after resolve | end | then: value
P6 return promise	x1 | y1 | y2 | y3 | x2 (inner) | y4
P7 timer delays	0ms first | 1ms | 0ms second | 20ms

HeadlessChrome/154.0.0.0

Figure — the ordering puzzles run in Chrome 154 (screenshot of the page after the script finished).

The explanations

Puzzle	Why
P1	`A`, `E` are synchronous. `C` and `D` are microtasks, run in the order queued. `B` is a task, so it waits for the microtask queue to empty.
P2	Calling an async function runs its body synchronously **up to the first** `await`. The rest is a microtask, so `2` prints before `f after await`.
P3	First the script's microtask `M1` runs; it queues a timer (behind T1 and T2) and a microtask, which runs immediately in the same checkpoint. Then task T1 runs; its microtask runs **before** task T2. Finally the timer queued by M1.
P4	Each `.then` in a chain is only queued when the previous one settles, so two chains alternate: a1, b1, a2, b2…
P5	The executor passed to `new Promise` runs synchronously. `resolve()` doesn't return or throw — later lines still run. `then` callbacks are always async, even on an already-resolved promise.
P6	When a `then` callback returns a promise (a **thenable**), the outer promise must *adopt* it: one microtask to call `inner.then(...)`, another for that reaction to fire, then the next `then`. So `x2` lands after `y3`. Returning a plain value would have printed `x2` right after `y1`.
P7	Timers fire in order of due time; equal due times in creation order. Node (and current Chrome) treat `0` like `1` ms here, so `1ms` sits between the two `0ms` timers — don't write code that depends on such fine differences.

P6 is why `return await somePromise` inside an `async` function used to be discouraged "for performance"; the difference is a couple of microtasks and doesn't matter in practice — and `return await` inside `try` is actually *required* to catch the rejection (§12.9).

In the interview

> **30-second answer:** "Synchronous code runs first. Then the engine drains the whole microtask queue — promise callbacks, code after `await`, `queueMicrotask` — including microtasks added along the way. Only then does the event loop take the next task, like a `setTimeout` callback or an event, and after each task it drains microtasks again. So promises always beat `setTimeout(0)`."

- *Q: Does `await` of a non-promise pause?* A: Yes — `await 5` still defers the rest of the function to a microtask.
- *Q: Why does P6's `x2` come late?* A: Adopting a returned thenable takes two extra microtask ticks.
- *Q: Can a microtask starve rendering?* A: Yes; the loop can't render or run tasks until the microtask queue is empty.

Self-check

1. `setTimeout(a, 0); queueMicrotask(b); Promise.resolve().then(c);` → order? → `b`, `c`, `a`.
2. What prints first: code inside `new Promise(executor)` or code after the `new Promise(...)` line? → The executor (it's synchronous).
3. A microtask queues a timer and another microtask. Which runs first? → The microtask.

TL;DR

Sync → all microtasks (FIFO, including new ones) → one task → all microtasks → … Executors and code before the first `await` are synchronous.

12.4 Node's event loop: `nextTick`, `setImmediate` and phases

What it is

Node's loop is implemented by **libuv** and cycles through **phases**, each with its own queue:

Phase	Runs
timers	Expired `setTimeout` / `setInterval` callbacks
pending callbacks	Some deferred system-level callbacks (e.g. certain TCP — the protocol under most network connections — errors)
poll	Waits for and runs I/O callbacks (files, sockets — a program's end of a network connection)
check	`setImmediate` callbacks
close callbacks	`'close'` events, e.g. `socket.on('close')`

Between every callback, Node drains two queues: first the **`process.nextTick` queue**, then the **promise microtask queue** (that's the order in CommonJS — Node's original `require`/`module.exports` module system; see the ES-module twist below — ES modules are the standard `import`/`export` system, also called ESM).

`code/03_node_queues.mjs`:

```
import { readFile } from 'node:fs';

const log = [];
setTimeout(() => log.push('timeout'), 0);
setImmediate(() => log.push('immediate'));
Promise.resolve().then(() => log.push('promise'));
process.nextTick(() => log.push('nextTick'));
log.push('sync');
setTimeout(() => {
  console.log('main module :', log.join(' | '));
  console.log('  (timeout vs immediate at top level can swap between runs - see text)');

  // Inside an I/O callback the order of immediate vs timeout IS guaranteed.
  readFile(new URL(import.meta.url), () => {
    const io = [];
    setTimeout(() => io.push('timeout'), 0);
    setImmediate(() => io.push('immediate'));
    process.nextTick(() => io.push('nextTick'));
    Promise.resolve().then(() => io.push('promise'));
    setTimeout(() => console.log('in I/O cb  :', io.join(' | ')), 20);
  });
}, 30);
```

Output:

```
main module : sync | promise | nextTick | immediate | timeout
  (timeout vs immediate at top level can swap between runs - see text)
in I/O cb  : nextTick | promise | immediate | timeout
```

And the CommonJS version, `code/03b_node_queues_cjs.js`:

```
const log = [];
Promise.resolve().then(() => log.push('promise'));
process.nextTick(() => log.push('nextTick'));
log.push('sync');
setTimeout(() => console.log('CommonJS main:', log.join(' | ')), 10);
```

Output:

```
CommonJS main: sync | nextTick | promise
```

Reading the results

1. **`nextTick` vs promises.** Normally `nextTick` callbacks run before promise callbacks (the I/O callback and the CommonJS script show this). But an ES module's top level is itself evaluated inside promise machinery, so its promise microtasks drain before Node gets to the `nextTick` queue — hence `promise | nextTick` in the `.mjs` main module. Don't build logic on this difference.
2. **`setImmediate` vs `setTimeout(0)` at top level** is **not deterministic** (not guaranteed to give the same result every run): it depends on whether 1 ms has already passed when the loop first checks timers. This run printed `immediate` first; another run may not.
3. **Inside an I/O callback** it *is* deterministic: after the poll phase comes the check phase, so `setImmediate` always wins over a new `setTimeout(0)`.

When to use which

Want	Use
Run after current code, before anything else (portable)	`queueMicrotask(fn)` or `Promise.resolve().then(fn)`
Node-only "right after this operation", e.g. emit an event after the constructor returns	`process.nextTick(fn)` (Node docs recommend `queueMicrotask` for most new code)
Yield to I/O, then continue	`setImmediate(fn)`
Run after a delay	`setTimeout(fn, ms)`

Common mistake: recursive `process.nextTick` starves I/O exactly like recursive microtasks do. The architecture of Node (libuv, thread pool — a fixed set of helper threads that do blocking work such as file reads) is covered in Chapter 15.

TL;DR

Node phases: timers → pending → poll (I/O) → check (`setImmediate`) → close. Between callbacks: `nextTick` queue, then promises (ESM top level flips these). `setImmediate` vs `setTimeout(0)` is only guaranteed inside I/O callbacks.

12.5 Timers

What it is

Function	Does	Cancel with
`setTimeout(fn, ms, ...args)`	Run `fn(...args)` once, **at least** `ms` later	`clearTimeout(id)`
`setInterval(fn, ms, ...args)`	Run repeatedly, roughly every `ms`	`clearInterval(id)`
`queueMicrotask(fn)`	Run as a microtask	—
`requestAnimationFrame(fn)` (browser)	Run before the next paint (Ch. 14)	`cancelAnimationFrame(id)`
`setImmediate(fn)` (Node)	Run in the check phase	`clearImmediate(id)`

The returned **handle** is a number in browsers and a `Timeout` object in Node (which also has `.unref()` to let the process exit while the timer is pending, and `.refresh()`).

How it works

1. The timer is registered with the host; your code continues.
2. When the delay has elapsed, the callback is put in the task queue.
3. It runs when the loop reaches it — possibly much later if the thread is busy.

The delay is therefore a **minimum**, never a guarantee. Other rules (HTML spec + practice):

- Negative delays become `0`.
- After 5 levels of nested timers (a timer scheduling a timer scheduling…), browsers clamp delays below 4 ms up to **4 ms**.
- Browsers throttle (deliberately slow down) timers in background tabs (often to once per second or less) to save battery.
- Delays above $2^{31}-1$ ms (about 24.8 days) overflow (exceed the largest number a 32-bit integer slot can hold); Node prints a warning and uses 1 ms.

`code/04_timers.mjs`:

```js
import { setTimeout as sleep } from 'node:timers/promises';

// 1) Extra arguments are passed to the callback; the return value is a handle
const t = setTimeout((a, b) => console.log('never runs', a, b), 50, 'x', 'y');
clearTimeout(t);
console.log('cleared a timer before it fired');

// 2) setInterval until a condition, then clearInterval
let ticks = 0;
const id = setInterval(() => {
  ticks += 1;
  console.log('tick', ticks);
  if (ticks === 3) clearInterval(id);
}, 10);
await sleep(60);

// 3) The delay is a MINIMUM: a busy main thread delays the callback
const t0 = performance.now();
setTimeout(() => console.log(`asked 10 ms, got >= 40 ms: ${performance.now() - t0 >= 40}`), 10);
while (performance.now() - t0 < 40) {}
await sleep(20);

// 4) Recursive setTimeout: the gap is measured from the END of each run
let n = 0;
function poll() {
  n += 1;
  if (n < 3) setTimeout(poll, 5);
  else console.log('recursive setTimeout ran', n, 'times');
}
poll();
await sleep(40);

// 5) Too-large delays overflow: > 2^31-1 ms (~24.8 days) becomes 1 ms
const big = setTimeout(() => console.log('huge delay fired immediately!'), 2 ** 31);
await sleep(20);
clearTimeout(big);

// 6) Promise-based sleep
const s = performance.now();
await sleep(15);
console.log(`awaited sleep(15): >= 15 ms: ${performance.now() - s >= 14.5}`);
```

Output (the three `(node:…)` lines are the warning Node writes to stderr):

```
cleared a timer before it fired
tick 1
tick 2
tick 3
asked 10 ms, got >= 40 ms: true
recursive setTimeout ran 3 times
(node:51319) TimeoutOverflowWarning: 2147483648 does not fit into a 32-bit signed integer.
Timeout duration was set to 1.
(Use `node --trace-warnings ...` to show where the warning was created)
huge delay fired immediately!
awaited sleep(15): >= 15 ms: true
```

`setInterval` vs recursive `setTimeout`

	`setInterval(fn, 100)`	recursive `setTimeout(fn, 100)`
Spacing	Start-to-start; if `fn` takes 80 ms only ~20 ms idle	End-to-start; always ≥ 100 ms of idle between runs
Slow or async work	Runs can bunch up or overlap (an async `fn` isn't awaited)	Next run scheduled only after this one finishes
Change the delay each time (backoff)	Awkward	Easy

For polling a server (asking it again and again whether something changed), prefer recursive `setTimeout` (after the previous request completes).

Common mistakes

- Passing the *result* of a call: `setTimeout(save(), 1000)` runs `save` now. Pass a function: `setTimeout(save, 1000)` or `() => save()`.
- Forgetting to clear intervals (e.g. when a component unmounts — is removed from the page in a UI framework such as React) → memory leaks (memory that can never be freed) and duplicate work.
- Using timers to "wait until something is ready" — listen for the actual event or await the actual promise instead.
- A method passed as callback loses `this`: `setTimeout(obj.method, 10)` — use an arrow or `bind` (Chapter 9).

TL;DR

Timer delays are minimums; the callback runs as a task when the thread is free. Clear what you schedule. Prefer recursive `setTimeout` for polling; `node:timers/promises` gives an awaitable `setTimeout`.

12.6 Callbacks and callback hell

What it is

A **callback** is a function you pass to another function, to be called later with the result. It's the oldest async style in JavaScript (event handlers, `setTimeout`, Node's original APIs). Node standardized **error-first callbacks**: `callback(err, result)` — `err` is `null` on success.

`code/05_callbacks.mjs` — key parts:

```
// 1) Error-first callbacks (the Node convention): cb(err, result)
readFile(new URL('./missing.txt', import.meta.url), 'utf8', (err, text) => {
  if (err) { console.log('error-first:', err.code); return; }
  console.log(text);
});

// 2) Callback hell: each step needs the previous result
db.getUser(1, (err, user) => {
  if (err) return console.log(err.message);
  db.getOrders(user, (err, orders) => {
    if (err) return console.log(err.message);
    db.getItems(orders[0], (err, items) => {
      if (err) return console.log(err.message);
      console.log('pyramid result:', user.name, orders[0].id, items);
    });
  });
});

// 3) Inversion of control: a buggy library calls your callback twice
function buggyCharge(amount, cb) { cb(null, 'ok'); cb(null, 'ok'); }
let charges = 0;
buggyCharge(10, () => { charges += 1; });
console.log('callback-style charges:', charges);

// A promise can settle only once, so the second call is ignored
const charged = new Promise((resolve) => buggyCharge(10, (_e, r) => resolve(r)));
let promiseCharges = 0;
await charged.then(() => { promiseCharges += 1; });
console.log('promise-style charges :', promiseCharges);

// 4) Zalgo: a callback that is sometimes sync, sometimes async
function getValue(key, cb) {
  if (cache.has(key)) cb(cache.get(key));          // sync!
  else setTimeout(() => cb(42), 0);                // async
}
// ...
```

Output:

```
callback-style charges: 2
promise-style charges : 1
error-first: ENOENT
pyramid result: Ada 7 [ 'book', 'pen' ]
zalgo order: cached cb | after call (a) | after call (b) | fetched cb
```

The problems callbacks have

Problem	What goes wrong	How promises fix it
Callback hell ("pyramid of doom")	Each dependent step nests deeper; error checks repeat at every level	Flat `.then` chains or linear `await` code
Error handling	`try/catch` can't catch errors thrown in a later callback; every level must check `err`	One `.catch` / one `try` catches errors from any step
Inversion of control	You trust someone else's code to call you once, not zero or twice	A promise settles exactly once
"Releasing Zalgo"	A callback that's sometimes sync and sometimes async makes ordering unpredictable (see the output: the cached callback ran *before* "after call")	`then` callbacks are always async
Composition	Running things in parallel and waiting for all is hand-written counting code	`Promise.all` and friends

Callbacks remain perfect for **repeated events** (clicks, stream data — a stream delivers data in chunks over time) — a promise can only deliver one value. Named functions and early `return`s flatten a pyramid a bit, but the real fix is promises.

TL;DR

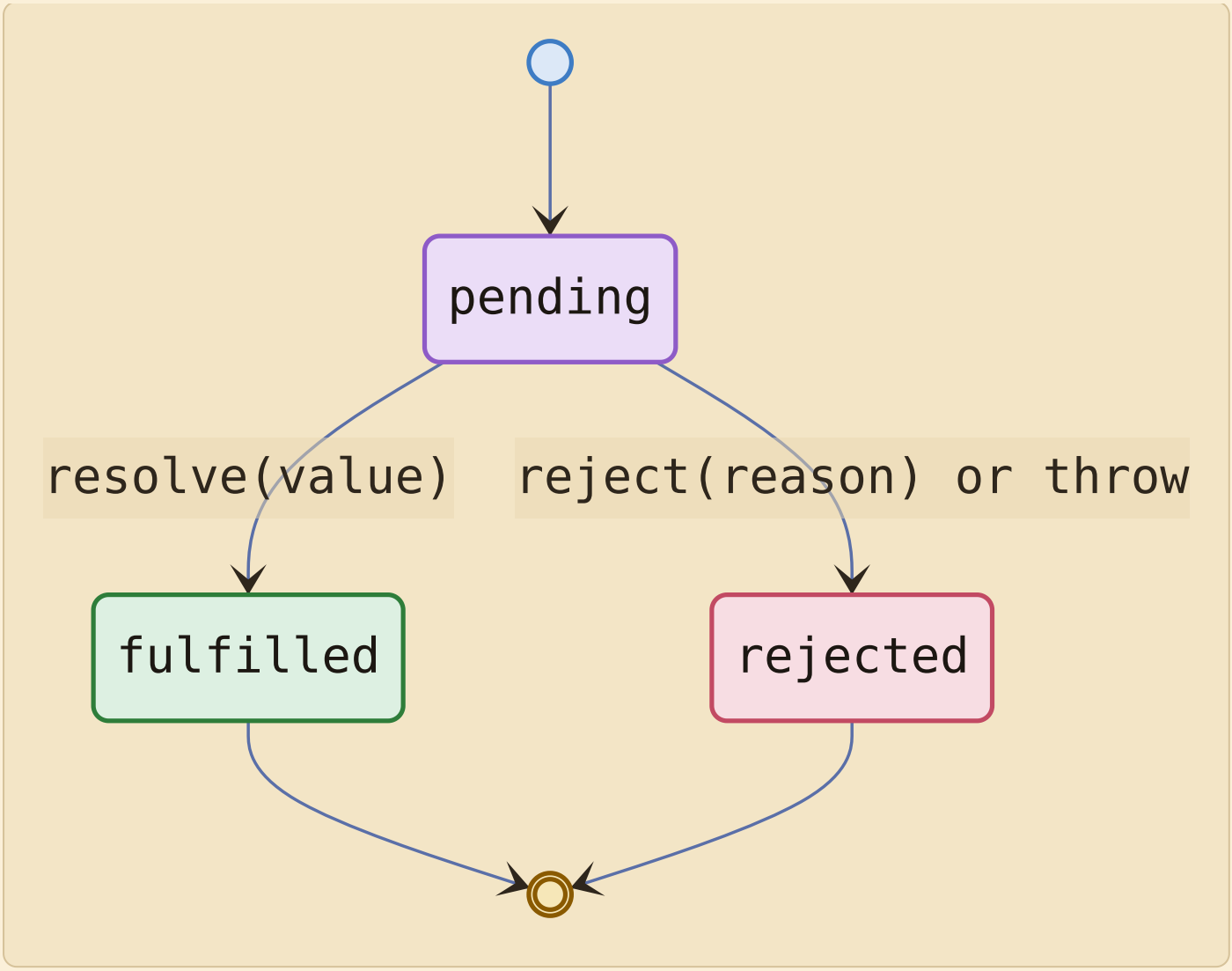
Callbacks: pass a function to be called later; Node uses `cb(err, result)`. They nest badly, can't use `try/catch`, and depend on trusting the caller — promises solve all of these. Keep callbacks for multi-fire events.

12.7 Promises

What it is

A **promise** is an object representing a value that will be available later — or an error explaining why not. Analogy: a pager a restaurant hands you. You can go sit down (your code continues); when the pager buzzes you either get your food (fulfilled) or an apology (rejected). It buzzes exactly once.

States:



- **pending** — not finished yet.
- **fulfilled** — finished with a value.
- **rejected** — failed with a reason (normally an `Error`).
- **settled** — fulfilled or rejected; a settled promise never changes again.
- **resolved** — "locked in": either settled, or following another promise. (A promise resolved with a pending promise is still pending.)

How it works

1. `new Promise((resolve, reject) => { … })` — the **executor** runs immediately and synchronously; it starts the work and later calls `resolve` or `reject`. If it throws, the promise is rejected.
2. `promise.then(onFulfilled, onRejected)` registers reactions and returns a **new** promise.
3. When the promise settles, reactions are queued as microtasks.
4. The new promise from `then` settles with whatever the reaction **returns** (a value → fulfilled; a promise → adopt its outcome) or **throws** (→ rejected).
5. A rejection skips `then(onFulfilled)` reactions until it meets a `catch` (or a `then` with `onRejected`). After `catch` returns normally, the chain is fulfilled again.
6. `finally(fn)` runs `fn` with no argument either way and passes the original outcome through (unless `fn` throws or returns a rejected promise).

`code/06_promises.mjs`:

```
const later = (value, ms, fail = false) =>
  new Promise((resolve, reject) => setTimeout(() => (fail ? reject(new Error(value)) :
resolve(value)), ms));

const p = later('done', 10);
console.log('right away :', p);      // Node shows the state
await p;
console.log('after await:', p);

// Chaining: each then() returns a NEW promise
const result = await Promise.resolve(2)
  .then((x) => x * 10)               // return a value -> next then gets it
  .then((x) => later(x + 1, 5))      // return a promise -> chain waits for it
  .then((x) => { throw new Error(`boom at ${x}`); })
  .then(() => console.log('skipped because the chain is rejected'))
  .catch((err) => { console.log('caught:', err.message); return 'recovered'; })
  .finally(() => console.log('finally: runs either way, sees no value'));
console.log('chain result:', result);

// Mistake: forgetting to return inside then() breaks the chain
await Promise.resolve(1)
  .then((x) => { later(x + 1, 5); })  // no return!
  .then((v) => console.log('forgot return ->', v));

// resolve/reject only count once
const once = new Promise((resolve, reject) => { resolve('first'); resolve('second'); reject(new
Error('ignored')); });
console.log('settles once:', await once);

// Promise.resolve adopts thenables; a promise is passed through unchanged
const thenable = { then(onOk) { onOk('from a thenable'); } };
console.log('thenable   :', await Promise.resolve(thenable));
console.log('same obj   :', Promise.resolve(p) === p);

// Promisify a callback API
const readFileP = promisify(readFile);
const text = await readFileP(new URL(import.meta.url), 'utf8');
console.log('promisify  :', text.split('\n')[0]);

// Promise.withResolvers (ES2024): resolve from outside the executor
const { promise, resolve } = Promise.withResolvers();
setTimeout(() => resolve('resolved from outside'), 5);
console.log('withResolvers:', await promise);

// Promise.try (ES2025): run a function that may throw synchronously OR return a promise
const risky = (x) => { if (x < 0) throw new RangeError('negative'); return later(x * 2, 1); };
for (const x of [3, -1]) {
  await Promise.try(risky, x).then((v) => console.log('Promise.try ok  :', v), (e) =>
console.log('Promise.try fail:', e.message));
}
```

Output:

```
right away : Promise { <pending> }
after await: Promise { 'done' }
caught: boom at 21
finally: runs either way, sees no value
chain result: recovered
forgot return -> undefined
settles once: first
thenable   : from a thenable
same obj   : true
promisify  : import { promisify } from 'node:util';
withResolvers: resolved from outside
Promise.try ok  : 6
Promise.try fail: negative
```

Key API

API	Meaning
`new Promise(executor)`	Wrap a callback API. Rarely needed otherwise (**explicit-construction anti-pattern** — an anti-pattern is a common habit that causes problems: wrapping something that already returns a promise).
`Promise.resolve(v)` / `Promise.reject(e)`	Already-settled promise (`resolve` adopts thenables, returns `v` itself if it's a native promise)
`.then(ok, fail)` / `.catch(fail)` / `.finally(fn)`	Reactions; each returns a new promise
`Promise.withResolvers()` (ES2024, the 2024 edition of the JS standard)	`{ promise, resolve, reject }` — handy when resolve is triggered by an event elsewhere
`Promise.try(fn, ...args)` (ES2025)	Call `fn`; a sync throw becomes a rejection, a returned value/promise is adopted
`util.promisify(fn)` (Node)	Turn an error-first callback function into a promise-returning one. Most Node APIs also have promise versions: `node:fs/promises`, `node:timers/promises`.

A **thenable** is any object with a `then` method; promises from different libraries interoperate through this duck-typed protocol (judged by which methods an object has, not by its class) (Chapter 9 — duck typing).

Common mistakes

- **Forgetting `return`** in a `then` callback (see output: `undefined`, and the inner promise's errors become unhandled).
- **Nesting `then`** inside `then` instead of returning — recreates callback hell.
- **`.then(ok, fail)` vs `.then(ok).catch(fail)`:** the first form does **not** catch errors thrown inside `ok`; the second does.
- **No `catch` at the end** → an unhandled rejection (crashes Node, logs in browsers) — §11.5.
- **Thinking promises are cancellable or lazy** (lazy = not starting until someone asks for the result). A promise represents work that has *already started*; use `AbortController` (§12.11) to cancel the work.

Self-check

1. What does `.then(() => 5)` return? → A new promise fulfilled with 5.
2. `Promise.resolve().then(() => { throw new Error('x') }).then(() => 1).catch(e => 2)` → fulfils with? → `2`.

3. What does `finally`'s callback receive? → Nothing; it can't see the value or reason.
4. Is the executor synchronous? → Yes.

TL;DR

A promise is a one-time, eventually-settled value: pending → fulfilled/rejected. `then` returns a new promise whose outcome is the callback's return value or thrown error. Always return inside `then`, end chains with `catch`.

12.8 Promise combinators: `all`, `allSettled`, `race`, `any`

What it is

Static methods (called on `Promise` itself, not on a promise instance) that take an iterable (anything `for…of` can loop over, such as an array) of promises (or plain values) and return one promise.

Method	Fulfils when	Rejects when	Result	Added
`Promise.all`	**all** fulfil	**first** rejection (fail-fast)	array of values in **input order**	ES2015
`Promise.allSettled`	all settle (never rejects)	never	array of `{status, value}` / `{status, reason}`	ES2020
`Promise.race`	first to **settle** is fulfilled	first to settle is rejected	that value/reason	ES2015
`Promise.any`	first to **fulfil**	**all** reject	value, or `AggregateError` with `.errors`	ES2021

`code/07_combinators.mjs`:

```
console.log('all       :', await Promise.all([later('slow', 30), later('fast', 10), 42]), at());
t0 = performance.now();
await Promise.all([later('ok', 30), later('db down', 10, true)])
  .catch((e) => console.log('all fails :', e.message, 'after', at()));

const settled = await Promise.allSettled([later('ok', 5), later('bad', 5, true)]);
console.log('allSettled:', settled.map((r) => (r.status === 'fulfilled' ? `ok:${r.value}` :
`err:${r.reason.message}`)));

console.log('race      :', await Promise.race([later('A', 20), later('B', 10)]));
await Promise.race([later('A', 20), later('B failed', 10, true)]).catch((e) => console.log('race err
:', e.message));

console.log('any       :', await Promise.any([later('mirror1 down', 5, true), later('mirror2', 15)]));
await Promise.any([later('x', 5, true), later('y', 5, true)])
  .catch((e) => console.log('any fails :', e.name, e.errors.map((x) => x.message)));

console.log('empty     :', await Promise.all([]), (await Promise.allSettled([])).length);
await Promise.any([]).catch((e) => console.log('any([])   :', e.name));
console.log('race([]) stays pending forever - never await it');

const withTimeout = (promise, ms) =>
  Promise.race([promise, new Promise((_, rej) => setTimeout(() => rej(new Error(`timed out after ${ms}
ms`)), ms))]);
await withTimeout(later('slow api', 100), 20).catch((e) => console.log('timeout   :', e.message));
```

Output:

```
all       : [ 'slow', 'fast', 42 ] 30 ms
all fails : db down after 10 ms
allSettled: [ 'ok:ok', 'err:bad' ]
race      : B
race err  : B failed
any       : mirror2
any fails : AggregateError [ 'x', 'y' ]
empty     : [] 0
any([])   : AggregateError
race([]) stays pending forever - never await it
timeout   : timed out after 20 ms
```

Notes:

- `all` took 30 ms (the slowest), not 40 — the promises ran concurrently (their waiting overlapped in time). The results are in input order even though `fast` finished first. Plain values like `42` are allowed.
- On failure, `all` rejects after 10 ms, but the other operation **keeps running** — combinators don't cancel anything. Pair them with an `AbortSignal` (§12.11) if you need that.
- `withTimeout` via `race` is a classic interview answer; in real code prefer `AbortSignal.timeout(ms)`, which also stops the underlying work (e.g. the HTTP request).

Choosing

Situation	Use
Load user, settings and feed; the page needs all three	`all`
Send 20 notifications; report which failed	`allSettled`
Fastest of several mirrors / CDNs (content delivery networks — servers worldwide holding copies of files)	`any`
A result or a timeout, whichever comes first	`race` (or `AbortSignal.timeout`)

In the interview

30-second answer: "`all` waits for everything and fails fast on the first rejection; `allSettled` waits for everything and never rejects; `race` settles with the first promise to settle, success or failure; `any` takes the first success and only rejects — with an `AggregateError` — if all fail. None of them cancel the losers."

- *Q: Implement `Promise.all`.* A: Return a new promise; keep a results array and a counter; for each input, `Promise.resolve(item).then(v => { results[i] = v; if (++done === n) resolve(results) }, reject)`; resolve immediately if the input is empty. (Chapter 17 has runnable polyfills — a polyfill is code that implements a built-in feature yourself, for environments that lack it.)
- *Q: `Promise.all` with one rejection — do the others stop?* A: No; their results are simply ignored.

TL;DR

`all` (all or first failure), `allSettled` (everything, never rejects), `race` (first settled), `any` (first success, else `AggregateError`). They don't cancel.

12.9 `async` / `await`

What it is

Syntax (ES2017) that lets you write promise-based code as if it were synchronous.

- An **`async function`** always returns a promise: a `return v` fulfils it with `v`; a `throw` rejects it.
- **`await expr`** pauses the async function until the promise settles, then gives back its value — or **throws** its rejection reason. Pausing doesn't block the thread: the function's remaining code is resumed later as a microtask, and meanwhile the caller continues.
- `await` is allowed only inside async functions and at the **top level of modules** (Chapter 10).

Under the hood an async function is like a generator (a function that can pause at `yield` and resume later — Chapter 8) that yields promises, driven by a runner that resumes it when each promise settles.

`code/08_async_await.mjs`:

```
// 1) An async function ALWAYS returns a promise
async function answer() { return 42; }
console.log('returns   :', answer() instanceof Promise, await answer());
async function fails() { throw new Error('nope'); }
console.log('throw ->  :', await fails().catch((e) => `rejected (${e.message})`));

// 2) Sequential vs parallel
let t = performance.now();
const a = await later('a', 30);
const b = await later('b', 30);
console.log('sequential:', a + b, round(performance.now() - t), 'ms');

t = performance.now();
const [c, d] = await Promise.all([later('c', 30), later('d', 30)]);
console.log('parallel  :', c + d, round(performance.now() - t), 'ms');

// 3) try/catch/finally with await
async function load(id) {
  try {
    return await later(`user ${id}`, 5, id < 0);   // "return await" keeps the error inside this try
  } catch (err) {
    return `fallback for ${err.message}`;
  } finally {
    console.log('  cleanup for', id);
  }
}
console.log('load(1)   :', await load(1));
console.log('load(-1)  :', await load(-1));

// 4) await in loops: for...of is sequential; forEach does NOT wait
const ids = [1, 2, 3];
const seen = [];
ids.forEach(async (id) => { await later(id, 12 - id * 3); seen.push(id); });
console.log('forEach done? seen =', seen, '(nothing yet!)');
await later(null, 20);
console.log('...later   seen =', seen, '(finish order, not input order)');
const inOrder = [];
for (const id of ids) inOrder.push(await later(id * 10, 2));
console.log('for...of  :', inOrder);
console.log('map+all   :', await Promise.all(ids.map((id) => later(id * 100, 5))));

// 5) Async arrow, async method, await of a non-promise
const double = async (x) => x * 2;
const obj = { async get() { return 'method'; } };
console.log('misc      :', await double(21), await obj.get(), await 'plain value');
```

Output:

```
returns   : true 42
throw ->  : rejected (nope)
sequential: ab 60 ms
parallel  : cd 30 ms
  cleanup for 1
load(1)   : user 1
  cleanup for -1
load(-1)  : fallback for user -1
forEach done? seen = [] (nothing yet!)
...later   seen = [ 3, 2, 1 ] (finish order, not input order)
for...of  : [ 10, 20, 30 ]
map+all   : [ 100, 200, 300 ]
misc      : 42 method plain value
```

Sequential vs parallel — the most common performance bug

Two independent `await`s in a row take the **sum** of their times (60 ms); starting both first and awaiting `Promise.all` takes the **max** (30 ms). Rule: `await` one after another only when step 2 needs step 1's result.

```
// Also parallel: start first, await later
const userP = getUser(id), postsP = getPosts(id);   // both requests in flight
const user = await userP, posts = await postsP;    // careful: if postsP rejects before
                                                    // userP settles, it's briefly "unhandled"
```

`Promise.all` is safer than the start-then-await form because it attaches handlers to every promise immediately.

Loops

Loop	Behaviour
`for…of` / `for` / `while` with `await` inside	One at a time, in order
`array.forEach(async …)`	Fires all, **doesn't wait**, ignores results and errors — avoid
`await Promise.all(array.map(async …))`	All at once, results in order
A concurrency-limited pool (§12.13)	At most N at a time

Error handling

- Wrap `await` in `try`/`catch`; the rejection reason arrives as the caught value.
- `return await p` inside `try` — without `await`, the function hands the promise back *before* it rejects, so your `catch` never sees the rejection and `finally` runs while the work is still in progress. Outside `try`, `return p` and `return await p` behave the same.
- Handle errors at the level that can do something about them (show a message, retry, fall back); let others propagate. Custom error classes and `cause` chaining are in Chapter 11.
- A promise you create but never await can reject with nobody listening → `unhandledrejection`.

Trade-offs: `async/await` vs `.then` chains

	`async` / `await`	`.then` chains
Readability	Reads like sync code; normal `if`, loops, `try`	Fine for short linear pipelines
Errors	`try` / `catch`	`.catch`
Parallelism	Must remember `Promise.all`	Same
Where it works	Async functions and module top level	Anywhere

They're the same machinery; mix freely (e.g. `await fetch(url).then(r => r.json())`).

In the interview

> **30-second answer:** "`async` functions always return promises; `await` suspends just that function until the promise settles, resuming it as a microtask, so the thread stays free. Rejections become exceptions, so `try/catch` works. The classic mistake is awaiting independent calls one by one — use `Promise.all` — and using `forEach` with an async callback, which doesn't wait."

- *Q: What does `async function f() {}; f()` return?* A: A promise fulfilled with `undefined`.
- *Q: Does code after an `await` run synchronously if the promise is already resolved?* A: No, it's always deferred to a microtask.
- *Q: Why `return await` inside `try`?* A: So the rejection is caught by that `try`'s `catch`.

Self-check

1. Three independent 100 ms requests awaited one after another take…? → ~300 ms; with `Promise.all` ~100 ms.
2. Can you use `await` in a regular function? → No (SyntaxError); only in async functions or module top level.
3. Why did `seen` end up `[3, 2, 1]`? → `forEach` started all three at once; they finished in order of their delays.

TL;DR

`async` → returns a promise; `await` → unwraps it or throws. Independent work → `Promise.all`. Sequential loop → `for…of`; never `forEach(async)`. `try { return await … } catch`.

12.10 Error handling across async code — a checklist

Chapter 11 introduced the rules (§11.5); now that you know the event loop, here is why they hold and how to apply them.

Situation	Where the error goes	Handle with
`throw` inside an `async` function	Rejects its returned promise	`try/catch` around `await f()`, or `f().catch()`
`throw` inside a `then` callback	Rejects the promise `then` returned	A later `.catch`
`throw` inside a `setTimeout` / event callback	Uncaught error in *that task* — no enclosing `try` exists any more	`try/catch` **inside** the callback; global `error` handler for logging
Rejection nobody handles	`unhandledrejection` event (browser) / crash (Node ≥ 15)	Always `await` / `return` / `.catch`
One of many parallel operations fails	`Promise.all` rejects with the first; others ignored	`allSettled` if you need every outcome
Error in an error-first callback	Passed as the first argument	`if (err) …` first thing

Rules of thumb:

1. Every promise has an owner who awaits it, returns it, or attaches `.catch`.
2. Catch where you can **act** (retry, fallback UI, HTTP 500 response); otherwise let it propagate, adding context with `new Error(msg, { cause })`.
3. Don't mix styles in one function: a callback API inside an async function should be promisified first, or its errors escape your `try`.
4. Global handlers (`window.onunhandledrejection`, `process.on('unhandledRejection')`) are for logging and reporting, not recovery.

TL;DR

try/catch works across `await` because rejections are re-thrown on resumption. Errors in plain callbacks belong to their own task. Leave no promise floating.

12.11 Cancellation: `AbortController` and `AbortSignal`

What it is

Promises can't be cancelled, but the *work* behind them can. The standard mechanism (from the DOM spec — the standard for the browser's page APIs — also in Node) is:

- **`AbortController`** — the object held by whoever decides to cancel; `controller.abort(reason?)`.
- **`AbortSignal`** — `controller.signal`, handed to the operation; it has `.aborted`, `.reason`, an `abort` event and `throwIfAborted()`.

Analogy: the controller is the "cancel order" button at the till; the signal is the ticket the kitchen watches.

API	Purpose	Support
`new AbortController()` / `.abort(reason)`	Manual cancel	All modern browsers, Node ≥ 15
`AbortSignal.timeout(ms)`	Signal that aborts with a `TimeoutError` `DOMException` (browser-API error type) after `ms`	Baseline (works in all major browsers) since 2022 (Node ≥ 17.3)
`AbortSignal.any([s1, s2])`	Aborts when any input aborts	Baseline since 2024 (Node ≥ 20.3)
`AbortSignal.abort(reason)`	Already-aborted signal	Modern browsers, Node
`{ signal }` option	Accepted by `fetch`, `addEventListener`, Node `timers/promises`, `fs.readFile`, streams, `events.on`…	—

`code/09_abort.mjs`:

```
// 1) Making your own function cancellable with an AbortSignal
function delay(ms, { signal } = {}) {
  return new Promise((resolve, reject) => {
    signal?.throwIfAborted();                       // already aborted? fail fast
    const id = setTimeout(() => { signal?.removeEventListener('abort', onAbort); resolve(`waited ${ms}
ms`); }, ms);
    function onAbort() { clearTimeout(id); reject(signal.reason); }
    signal?.addEventListener('abort', onAbort, { once: true });
  });
}

const controller = new AbortController();
setTimeout(() => controller.abort(), 20);
try {
  await delay(1000, { signal: controller.signal });
} catch (err) {
  console.log('manual abort  :', err.name, '-', err.message);
}
console.log('signal state  :', controller.signal.aborted);

// 2) Custom reason
const c2 = new AbortController();
c2.abort(new Error('user left the page'));
await delay(10, { signal: c2.signal }).catch((e) => console.log('custom reason :', e.message));

// 3) AbortSignal.timeout(ms) - aborts with a TimeoutError
await delay(1000, { signal: AbortSignal.timeout(15) }).catch((e) => console.log('timeout       :',
e.name));

// 4) AbortSignal.any - abort if ANY of several signals aborts
const userCancel = new AbortController();
const combined = AbortSignal.any([userCancel.signal, AbortSignal.timeout(5000)]);
setTimeout(() => userCancel.abort(), 10);
await delay(1000, { signal: combined }).catch((e) => console.log('any()         :', e.name));

// 5) Built-in APIs accept signals too
await sleep(1000, null, { signal: AbortSignal.timeout(10) }).catch((e) => console.log('node sleep
:', e.name));

const target = new EventTarget();
const listeners = new AbortController();
target.addEventListener('ping', () => console.log('ping heard'), { signal: listeners.signal });
target.dispatchEvent(new Event('ping'));
listeners.abort();                                  // removes the listener
target.dispatchEvent(new Event('ping'));
console.log('listener removed by abort()');
```

Output:

```
manual abort  : AbortError - This operation was aborted
signal state  : true
custom reason : user left the page
timeout       : TimeoutError
any()         : AbortError
node sleep    : AbortError
ping heard
listener removed by abort()
```

How to write abortable functions

1. Accept `{ signal }` in an options object.
2. Check `signal?.throwIfAborted()` at the start (and between steps of long work).
3. Listen for `abort` once; on abort, undo/stop the work and reject with `signal.reason`.
4. Remove the listener when the work finishes normally (avoid leaks on long-lived signals).
5. Pass the signal down to every API you call (`fetch(url, { signal })`).

Callers distinguish a deliberate cancel from a real failure by `err.name === 'AbortError'` (or `'TimeoutError'`) and usually ignore it silently. Fetch with abort and timeouts against a real server is in Chapter 14.

Real-world examples

- Search-as-you-type: abort the previous request when a new keystroke arrives.
- React effects (code React runs after a component appears on screen): abort the fetch in the effect's cleanup function when the component unmounts.
- Removing a group of event listeners at once: one controller, `{ signal }` on each `addEventListener`, one `abort()`.

TL;DR

`AbortController` cancels, `AbortSignal` carries the cancel to the work. `AbortSignal.timeout` and `AbortSignal.any` cover timeouts and combined cancel. Rejections have `name` `AbortError`/`TimeoutError`.

12.12 Async iteration: `for await…of` and async generators

What it is

Chapter 8 showed the (sync) **iterable protocol**: `[Symbol.iterator]()` returns an iterator whose `next()` returns `{ value, done }`. The **async iterable protocol** (ES2018) is the same, except the method is `[Symbol.asyncIterator]()` and `next()` returns a **promise** of `{ value, done }`. You consume it with **`for await (const x of source)`**, and you produce one easily with an **async generator** (`async function*`), which can both `await` and `yield`.

Use it for data that arrives in pieces over time: paginated APIs (APIs that return a big result one page at a time), file and network streams (Node streams and web `ReadableStream`s are async iterable), event streams.

`code/10_async_iteration.mjs`:

```
// 1) An async generator: yields values that arrive over time
async function* countdown(from, ms) {
  for (let i = from; i > 0; i--) {
    await new Promise((r) => setTimeout(r, ms));
    yield i;
  }
}
for await (const n of countdown(3, 5)) console.log('countdown', n);

// 2) Paginated API -> one flat async stream
const pages = { 1: { items: ['a', 'b'], next: 2 }, 2: { items: ['c'], next: 3 }, 3: { items: ['d',
'e'], next: null } };
const fetchPage = async (n) => { await null; return pages[n]; };
async function* allItems() {
  let page = 1;
  while (page) {
    const { items, next } = await fetchPage(page);
    yield* items;                    // delegate to the array's iterator
    page = next;
  }
}
const collected = [];
for await (const item of allItems()) {
  collected.push(item);
  if (collected.length === 4) break;   // break calls return() -> generator stops fetching
}
console.log('first 4 items :', collected);

// 3) Array.fromAsync (ES2026) collects an async iterable into an array
console.log('fromAsync     :', await Array.fromAsync(allItems()));

// 4) for await also works on an array of promises (awaits each in order)
const ps = [Promise.resolve('p1'), new Promise((r) => setTimeout(() => r('p2'), 5))];
for await (const v of ps) console.log('for await over promises:', v);

// 5) The protocol by hand
const ticker = {
  [Symbol.asyncIterator]() {
    let i = 0;
    return { next: async () => (i < 2 ? { value: `tick${i++}`, done: false } : { value: undefined,
done: true }) };
  },
};
for await (const tk of ticker) console.log('manual iterator:', tk);

// 6) Events as an async stream (Node events.on)
const bus = new EventEmitter();
const ac = new AbortController();
setTimeout(() => { bus.emit('msg', 'hello'); bus.emit('msg', 'world'); ac.abort(); }, 5);
try {
  for await (const [text] of on(bus, 'msg', { signal: ac.signal })) console.log('event:', text);
} catch (e) {
  console.log('stream ended  :', e.name);
}
```

Output:

```
countdown 3
countdown 2
countdown 1
first 4 items : [ 'a', 'b', 'c', 'd' ]
fromAsync     : [ 'a', 'b', 'c', 'd', 'e' ]
for await over promises: p1
for await over promises: p2
manual iterator: tick0
manual iterator: tick1
event: hello
event: world
stream ended  : AbortError
```

Notes

- `for await` is allowed wherever `await` is (async functions, module top level).
- `break`, `return` or a thrown error inside the loop calls the iterator's `return()`, so an async generator's `finally` blocks run — useful for closing connections.
- Items are processed **one at a time**. For an array of independent promises, `Promise.all` is usually better; `for await` over such an array also can't report a rejection until the loop reaches it.
- `Array.fromAsync` is part of ES2026 and supported in Node 22+ and current browsers.

> **TL;DR**

Async iterables deliver `{value, done}` promises; `for await…of` consumes them; `async function*` produces them. Great for pagination and streams; `break` cleans up.

12.13 Real-world async patterns

`code/11_patterns.mjs` implements three helpers that appear in almost every production codebase (code that runs the real, live service).

```
// 1) Retry with exponential backoff
async function retry(fn, { tries = 4, base = 5 } = {}) {
  for (let attempt = 1; ; attempt++) {
    try {
      return await fn(attempt);
    } catch (err) {
      if (attempt >= tries) throw new Error(`gave up after ${attempt} tries`, { cause: err });
      const wait = base * 2 ** (attempt - 1);   // 5, 10, 20 ... (add random jitter in production)
      console.log(`  attempt ${attempt} failed (${err.message}); retrying in ${wait} ms`);
      await sleep(wait);
    }
  }
}

// 2) Run many tasks but at most `limit` at once
async function mapLimit(items, limit, fn) {
  const results = new Array(items.length);
  let next = 0, running = 0, peak = 0;
  async function worker() {
    while (next < items.length) {
      const i = next++;                          // safe: no other code runs between read and write
      running++; peak = Math.max(peak, running);
      results[i] = await fn(items[i]);
      running--;
    }
  }
  await Promise.all(Array.from({ length: limit }, worker));
  return { results, peak };
}

// 3) Race condition: an older, slower response arrives last and overwrites newer data
let latest = 0;
await Promise.all(['ja', 'jav'].map(async (q) => {
  const ticket = ++latest;
  const r = await search(q);
  if (ticket === latest) shown = r;                // ignore answers to old requests
}));
```

Output:

```
  attempt 1 failed (503); retrying in 5 ms
  attempt 2 failed (503); retrying in 10 ms
retry result: payload
mapLimit    : [
   1,  4,  9, 16,
  25, 36, 49
] peak concurrency = 3
naive shows : results for "ja" <- stale!
guarded     : results for "jav"
```

Pattern	Problem it solves	Production notes
Retry + exponential backoff (double the wait after each failure)	Transient (temporary) failures (HTTP 503 "service unavailable", network blips)	Add random **jitter** so many clients don't retry in sync; retry only idempotent operations (safe to repeat — doing them twice equals doing them once; see System Design Fundamentals); cap attempts and total time; respect `Retry-After` headers
Concurrency limit	10,000 requests at once would overload the server or hit rate limits (caps on requests per second) / file-descriptor limits (caps on open files and sockets)	Libraries: `p-limit`, `p-map`; the worker-pool shape above is enough for scripts
Last-request-wins	Responses arrive out of order (search-as-you-type), stale data overwrites fresh — a **race condition** (the result depends on which operation happens to finish first)	Better still: abort the old request (§12.11); React Query and similar libraries do this for you
Timeout	A request that never answers	`AbortSignal.timeout(ms)`
Debounce / throttle (wait until events stop / run at most once per interval)	Too many events (keystrokes, scroll)	Implemented in Chapter 17

Note `mapLimit`'s `next++`: because of run-to-completion, no other worker can run between reading and incrementing `next`, so no lock is needed — a direct benefit of the single thread.

> **TL;DR**
>
> Retry with backoff + jitter for transient errors; cap concurrency with a worker pool; guard against stale responses with request IDs or abort; time out everything that talks to the network.

Chapter 12 TL;DR

- **One thread**, run-to-completion; the host does the waiting. Long sync work blocks timers, input and rendering.
- **Event loop:** one task → drain **all** microtasks → maybe render (rAF = `requestAnimationFrame` → style → layout → paint) → repeat. Promise reactions, code after `await`, `queueMicrotask` = microtasks; timers, I/O, events = tasks.
- **Puzzles:** sync first (including executors and code before the first `await`), then microtasks FIFO, then tasks. Returning a promise from `then` costs 2 extra ticks.
- **Node:** phases timers → poll → check (`setImmediate`) → close; `nextTick` before promises (except at ESM top level); `setImmediate` vs `setTimeout(0)` only ordered inside I/O callbacks.
- **Timers** are minimum delays; nested clamp 4 ms; overflow above $2^{31}-1$ ms; prefer recursive `setTimeout` for polling.
- **Callbacks** → callback hell, no `try/catch`, inversion of control, Zalgo. Keep them for repeated events.
- **Promises:** pending → fulfilled/rejected, settle once; `then` returns a new promise; always `return` and end with `catch`. `withResolvers` (ES2024), `Promise.try` (ES2025).
- **Combinators:** `all`, `allSettled`, `race`, `any` — none cancel.
- **async/await:** returns a promise; `await` throws rejections; use `Promise.all` for independent work; no `forEach(async)`; `return await` inside `try`.

- **Cancel** with `AbortController` / `AbortSignal.timeout` / `AbortSignal.any`.
- **Async iteration:** `for await…of`, `async function*`, `Array.fromAsync` (ES2026).
- **Patterns:** retry with backoff + jitter, concurrency limits, last-request-wins.

Next: the page itself — Chapter 13: DOM and Events.